

Forecasting the Duration of Price Instability Following Global Shocks:

A Literature Review

1. Introduction

Global shock events—pandemics, wars, and financial crises—have repeatedly destabilized financial markets with severity and speed that outpace conventional risk models. COVID-19 erased over 33% of the S&P 500's value within a single month, with volatility rivaling the Great Crash of 1929 and the GFC; U.S. circuit breakers were triggered four times in ten days, a frequency unprecedented since their 1987 introduction, while major European and Asian indices registered comparable losses (Baker et al., 2020, §1; Zhang et al., 2020, §1).

Such shocks induce a phase of abnormally elevated volatility—hereafter termed panic volatility—during which sentiment-driven trading dominates and prices decouple from fundamentals (Zhang et al., 2020, §3.1; Izzeldin et al., 2023, §1). The central question this paper addresses is: how long does panic volatility persist following a global shock before markets stabilize at a new equilibrium? The answer carries direct implications for hedging strategy design, policy intervention timing, and capital allocation under sustained uncertainty.

The practical stakes are illustrated by the agricultural commodity markets' response to the Russian invasion of Ukraine on February 24, 2022. Russia and Ukraine account for approximately 32% of global wheat and barley exports (Aliu et al., 2023, §1); the invasion triggered immediate supply disruptions and sharp price spikes, with wheat recording the largest proportional volatility increase of any commodity and agricultural markets remaining in elevated-volatility regimes roughly twice as long as during COVID-19 (Izzeldin et al., 2023, §4.2).

To address this gap, we apply machine learning as a *regression* tool to prospectively forecast crisis duration as a continuous target—an approach not attempted in the existing literature. Section 2 reviews three thematic blocks: (§2.1) market reactions to global shocks across equities and commodities; (§2.2) econometric methods for detecting structural breaks and market regimes; and (§2.3) machine learning applications in financial forecasting and their applicability to duration prediction.

2. Literature Review

2.1 Market Volatility Following Global Shocks

Global shocks do not affect financial markets uniformly. Evidence across the GFC, COVID-19, and the Ukraine war reveals systematic variation in how different asset classes respond, in terms of both severity and duration.

Equity Markets

COVID-19 produced the most severe and broad-based equity disruption of the three events: risk levels across twelve major economies nearly tripled between February and March 2020, driven primarily by sentiment and policy uncertainty rather than the direct health impact of the virus, with markets becoming highly synchronized following the WHO pandemic announcement (Zhang et al., 2020, §3.1; Izzeldin et al., 2021, §5). Sector responses diverged markedly—Oil and Gas entered a crisis regime first, Technology

last, as lockdown measures accelerated digital adoption (Izzeldin et al., 2021, §4). Earlier containment was associated with measurably lower crisis intensity (Izzeldin et al., 2021, Table 4). The Ukraine invasion produced a qualitatively different response: equity markets reacted instantaneously—unlike the multi-day lags observed after COVID-19 and the GFC—yet crisis duration and intensity for stock indices were considerably more muted (Izzeldin et al., 2023, §4.1; Table 3).

Commodity Markets

During COVID-19, oil bore the greatest commodity impact as global lockdowns caused an abrupt demand collapse, pushing crude futures to $-\$37.63/\text{barrel}$ on April 20, 2020 (Ji et al., 2020, §1). Agricultural markets responded with a lag of up to 17 days, while gold retained the lowest conditional volatility across all three crises (Izzeldin et al., 2023, Table 2; Table 4). The Ukraine war reversed this pattern: wheat prices rose 27% and sunflower oil 36.5% by May 2022, and agricultural commodities remained in elevated-volatility regimes for approximately twice as long as during COVID-19 (Aliu et al., 2023, §3.1; Izzeldin et al., 2023, §4.2).

Asset-Structural Roles

These patterns reflect distinct structural roles in the global economy. Oil is demand-driven and acutely sensitive to activity shocks (Li & Umair, 2023, §1). Gold functions as a safe-haven: across all three crises it showed the lowest spillover from oil risk shocks, remained decoupled from the broader commodity system under stress, and retained positive Sharpe ratios throughout COVID-19 (Umar et al., 2021b, §5.2; Ji et al., 2020, §4.2)—though safe-haven properties are not fixed and depend on crisis type (Ji et al., 2020, §5). Wheat is primarily supply-driven, with comparatively low volatility connectedness to oil price shocks but subject to a Granger-causal relationship through fertilizer, machinery, and transportation costs that intensifies during crises (Umar et al., 2021a, §5.1, §5.3). These structural differences justify treating oil, gold, wheat, and stock indices as distinct asset classes in the present study.

Crisis Duration: The Izzeldin et al. (2023) Precedent

The paper most directly relevant to the present study is Izzeldin et al. (2023), who apply a Markov-switching heterogeneous autoregressive (MS-HAR) model to daily volatility data across the GFC, COVID-19, and the Ukraine war, estimating synchronization, duration, and intensity for each asset-crisis pair. Duration is defined as total time spent in the crisis regime, expressed in days and as a percentage of the estimation window. Results show substantial variation: stock markets spent considerably less time in the crisis regime following Ukraine than during COVID-19 or the GFC; agricultural commodities showed approximately twice the median duration after Ukraine compared to COVID-19; and wheat recorded the highest crisis intensity at 48.87% (Table 3; Table 4; §4.2). Critically, this framework is retrospective—it quantifies how long past crises lasted once fully resolved. Our study addresses the complementary prospective question: forecasting duration at shock onset.

2.2 Methods for Identifying Market Phases and Structural Breaks

Forecasting instability duration requires a clear definition of when instability begins and ends. Three methodological frameworks are directly relevant.

Structural Break Detection

The Chow (1960) test assesses parameter stability across two subperiods but requires a pre-specified break date, limiting its utility for crisis detection. The Bai-Perron method (1998, 2003) removes this

constraint by endogenously detecting multiple breaks via global minimization of squared residuals, subject to a minimum distance constraint between breaks. It produces statistically valid confidence intervals, accommodates non-stationary series, and has become the standard tool in financial econometrics for identifying crisis onset dates, capital structure changes, and policy regime shifts (Bai & Perron, 1998, §2; 2003, §2). In our framework, Bai-Perron logic identifies the date at which volatility undergoes a permanent upward shift relative to pre-crisis baselines.

GARCH Family

Engle's (1982) ARCH and Bollerslev's (1986) GARCH model conditional variance as a function of past shocks and past variance: $\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$, where α_1 captures the ARCH effect and β_1 captures volatility persistence. High β_1 implies long-lasting volatility shocks, consistent with crisis regimes. Asymmetric extensions—EGARCH (Nelson, 1991, §2) and TGARCH (Zakoian, 1994, §2)—incorporate the leverage effect (Black, 1976): negative shocks raise future volatility disproportionately, an asymmetry that is particularly pronounced during crises and directly relevant to modelling instability intensity and persistence.

Regime-Switching Models

Hamilton (1989) introduced the Markov-switching framework, in which market states follow a hidden Markov chain with time-varying transition probabilities (§2; §3). Izzeldin et al. (2023) extend this via an MS-HAR model that allows volatility dynamics to differ between normal and crisis regimes, defining crisis end as the point at which the filtered probability of remaining in the crisis state falls below 50% (§3). Our study integrates all three lenses: Bai-Perron logic to identify shock onset, GARCH-derived features to characterize shock severity, and regime-switching logic to define the return to stability. The methodological novelty is mapping features from these classical frameworks into prospective duration predictions using machine learning.

2.3 Machine Learning for Volatility and Duration Forecasting

Performance Advantages in High-Uncertainty Regimes

Machine learning methods capture complex nonlinear patterns and feature interactions without parametric assumptions, and show particular performance advantages during crisis periods—exactly when accurate forecasts are most valuable. Chen and Guestrin (2016) introduced XGBoost as an efficient gradient-boosting implementation that has become the benchmark for tabular data forecasting, with regularization mechanisms and high-order interaction modelling well-suited to crisis prediction. Umar et al. (2021) and Sharma et al. (2021) find that SVMs outperform GARCH pre-crisis but degrade during COVID-19, whereas XGBoost and ensemble methods maintain stable performance across regimes. Awijen et al. (2023) confirm that machine learning significantly outperforms classical time-series methods for oil price forecasting under high uncertainty.

Duration as a Regression Target

Existing ML studies in finance address next-period volatility prediction: a valuable but insufficient formulation for strategic decision-making. Whether elevated volatility persists for 3, 20, or 60 more days carries fundamentally different implications for hedging and capital allocation. Izzeldin et al. (2023) address duration retrospectively; no existing work attempts prospective duration forecasting. Our

formulation—duration as a continuous regression target rather than a binary classification or next-period prediction—is the core methodological novelty of this study.

Network-Based Features

Xiu et al. (2021) demonstrate that SVD entropy of the stock correlation network (Almog & Shmueli, 2019) declines sharply during crashes as herding behavior erases firm-specific variation (§3; §4). We hypothesize that the rate at which network entropy recovers following a shock is informative about crisis duration: markets where correlation structure rapidly restabilizes may exit crisis regimes faster. Incorporating network topology metrics bridges traditional time-series forecasting and network analysis—a synergy that has received limited attention in the crisis-duration literature.

Candidate Algorithms

We implement and compare XGBoost, Random Forest, Support Vector Regression, and classical ARIMA/GARCH baselines. XGBoost’s regularization and interaction-modelling capacity make it the primary candidate; Random Forest offers interpretable feature importance rankings; SVR has shown promise in financial applications but requires careful kernel selection. Classical baselines are included as performance benchmarks. Algorithm selection is determined empirically on a held-out test set using cross-validated early stopping, rather than committed to a priori.

Research Gap and Contribution

The existing literature provides: (1) retrospective crisis duration measures (Izzeldin et al., 2023); (2) methods for detecting crisis onset but not forecasting crisis end-dates (Bai & Perron, 2003; Hamilton, 1989); (3) ML applications to next-period volatility but not duration (Awijen et al., 2023; Sharma et al., 2021); and (4) network analysis of structural crisis changes with limited forecasting application (Xiu et al., 2021). Our contribution is to unite these threads: prospective, machine-learning-enabled forecasting of crisis duration as a continuous regression target, integrating volatility, structural break, regime-switching, and network features. To our knowledge, this is the first study to address crisis duration forecasting in this integrated framework.

References

- Aliu, F., Caporale, G. M., & Spagnolo, N. (2023). Assessing the impact of the Russia-Ukraine war on food price inflation. *Journal of Policy Modeling*, 52, 10–25.
- Almog, A., & Shmueli, E. (2019). Structural entropy: Monitoring correlation-based networks over time with application to financial markets. *Scientific Reports*, 9, 10832.
- Awijen, H., Ben Ameer, H., Ftiti, Z., & Louhichi, W. (2023). Forecasting oil price in times of crisis: New evidence from machine learning versus deep learning models. *Annals of Operations Research*.
- Bai, J., & Perron, P. (1998). Estimating and testing linear models with multiple structural changes. *Econometrica*, 66(1), 47–78.
- Bai, J., & Perron, P. (2003). Computation and analysis of multiple structural change models. *Journal of Applied Econometrics*, 18(1), 1–22.
- Baker, S. R., Bloom, N., Davis, S. J., & Terry, S. J. (2020). COVID-induced economic uncertainty. NBER Working Paper No. 26983.
- Black, F. (1976). Studies of stock price volatility changes. *Proceedings of the 1976 American Statistical Association Meeting*, 177–181.
- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327.

- Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794.
- Chow, G. C. (1960). Tests of equality between sets of coefficients in two linear regressions. *Econometrica*, 28(3), 591–605.
- Engle, R. F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007.
- Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2), 357–384.
- Izzeldin, M., Muradoglu, Y., & Pappas, V. (2021). The impact of government and monetary policy responses to COVID-19 on stock market liquidity. *Journal of International Money and Finance*, 127, 102787.
- Izzeldin, M., Sohaiby, A., & Pappas, V. (2023). Measuring the duration of crisis episodes in financial markets: Evidence from the GFC, COVID-19, and the Russia-Ukraine war. *Journal of International Financial Markets, Institutions and Money*, 82, 101701.
- Ji, Q., Zhang, D., & Zhao, Y. (2020). Searching for safe-haven assets during the COVID-19 pandemic. *International Review of Financial Analysis*, 71, 101526.
- Li, Y., & Umair, M. (2023). Oil price shocks and their impact on economic growth: A structural VAR approach. *Energy Economics*, 117, 106460.
- Nelson, D. B. (1991). Conditional heteroskedasticity in asset returns: A new approach. *Econometrica*, 59(2), 347–370.
- Sharma, S., Aggarwal, V., & Yadav, M. P. (2021). Comparison of linear and non-linear GARCH models for forecasting volatility of select emerging countries. *Journal of Advances in Management Research*, 18(4), 526–547.
- Umar, M., Mirza, N., Rizvi, S. K. A., & Furqan, M. (2021a). Asymmetric volatility structure of equity returns: Evidence from an emerging market. *Quarterly Review of Economics and Finance*, 79, 120–135.
- Umar, M., Su, C. W., & Mirza, N. (2021b). Does geopolitical risk matter for global asset returns? Evidence from quantile regression. *Finance Research Letters*, 45, 102149.
- Xiu, Y., Wang, G., & Chan, W. K. V. (2021). Crash diagnosis and price rebound prediction in NYSE Composite Index based on visibility graph and time-evolving stock correlation network. *Entropy*, 23(12), 1612.
- Zakoian, J. M. (1994). Threshold heteroskedastic models. *Journal of Economic Dynamics and Control*, 18(5), 931–955.
- Zhang, D., Hu, M., & Ji, Q. (2020). Financial markets under the global pandemic of COVID-19. *Finance Research Letters*, 36, 101528.